

**PROJECT : SENTIMENT ANALYSIS OF NETFLIX
MOVIES AND SERIES USING NLP**

ABSTRACT

CineSense is a web-based platform for analyzing movies and series that helps users explore Netflix content and understand what audiences think through Natural Language Processing (NLP)-based sentiment analysis. The system provides details about movies and TV shows, including trending and popular content, ratings, popularity, descriptions, genres, release dates, watch-provider availability, and audience reviews. The application uses the TMDB API as an external source for movie and TV metadata and review data.

CineSense aims to improve traditional movie-rating systems by examining the opinions expressed in audience reviews. When a user opens a details page for a specific movie or series, the system retrieves the available reviews and processes them using a pretrained RoBERTa-based NLP sentiment model. It classifies the reviews into positive, neutral, or negative categories. The application then combines these results to produce an overall sentiment distribution for the chosen title. The system is built using Next.js for the web application and server-side processing, while Redis is suggested for caching previous sentiment-analysis results. Caching reduces repeated processing and speeds up response time when the same title is accessed multiple times. CineSense combines movie discovery, audience review analysis, and NLP-based sentiment classification to give users a deeper understanding of audience perception beyond just numerical ratings.

[KEY WORDS : Sentiment analysis classifier, RoBERTa-based NLP, TMDB API, User rating correlation **]**

1. INTRODUCTION

The rapid growth of digital streaming platforms has changed the way people discover, evaluate, and watch movies and television series. Platforms such as Netflix provide users with access to a huge and constantly changing collection of content. However, having so many choices can also make it difficult for users to decide what to watch. Traditionally, people rely on factors such as ratings, popularity, genre, descriptions, recommendations, and release dates when evaluating a movie or series. While these factors are useful, they do not always capture how viewers actually feel about the content. Audience reviews, on the other hand, contain detailed opinions about different aspects of a movie or series, such as acting, storyline, characters, pacing, cinematography, visual effects, and overall entertainment value. [1] [2]

CineSense is a web-based platform developed to combine movie and series discovery with Natural Language Processing (NLP)-based sentiment analysis. The main goal of the system is to help users explore movies and television series available on Netflix while also giving them a clearer idea of how audiences feel about those titles. Users can browse trending, popular, and highly rated movies and series. The platform also includes a search feature that allows users to find specific titles using keywords. After selecting a title, users can view detailed information such as ratings, popularity, overview, genres, release date, watch-provider availability, and audience reviews.

The application obtains movie and television-related information through the TMDB API. This provides the required metadata and review information without requiring CineSense to maintain a large movie database of its own. By using an external API, the system can focus mainly on presenting and analyzing the available information while avoiding the additional complexity of collecting and maintaining a large amount of entertainment data. One of the main features of CineSense is its sentiment-analysis system. The model classifies reviews into three categories: positive, neutral, and negative. [2]

To make the system more efficient, CineSense also uses Redis for caching previously generated sentiment-analysis results. Sentiment analysis involves additional processing, so analyzing the same reviews every time a user opens a particular title would be unnecessary. By storing previously generated results in Redis, the application can reuse them when the same movie or series is accessed again. This helps reduce repeated processing and can improve the overall response time of the application.

Overall, CineSense brings together web development, external API integration, Natural Language Processing, sentiment analysis, and caching within a single software engineering project. Instead of relying only on numerical ratings, the system uses audience reviews to provide additional insight into how viewers perceive movies and television series. In this way, CineSense aims to make the process of discovering and evaluating entertainment content more informative and user-friendly.

1.1 Purpose

The rapid growth of online streaming platforms has led to a large and ever-growing collection of movies and television series. Users often rely on numerical ratings, popularity scores, descriptions, and recommendations when choosing what to watch. While these metrics provide a general idea of a title, they do not always reflect the detailed opinions and experiences of viewers. Written reviews offer valuable insights into acting, storylines, characters, pacing, visual quality, and overall satisfaction, but reading many reviews manually can be time-consuming.

CineSense aims to provide a single platform that combines traditional movie and series information with automatic sentiment analysis of audience reviews. The system lets users find popular and trending Netflix content, search for specific titles, and view detailed information about selected movies and series. Additionally, CineSense uses a pretrained RoBERTa-based NLP model to classify audience reviews into positive, neutral, and negative categories. This sentiment distribution gives users a summarized view of audience perception. Therefore, the project aims to show how NLP can be integrated into a modern web application to improve how users interpret textual audience feedback.

1.2 Scope

CineSense focuses on discovering and analyzing movies and television series available on Netflix in the selected region. The system provides sections for trending, popular, and highly rated movies and series, along with a keyword-based search function. Users can open a detailed page for a chosen title and see information such as its title, poster, rating, popularity, overview, genres, release information, watch-provider availability, and audience reviews. The project uses the TMDB API as the main source of movie and television data.

The sentiment-analysis feature activates when a user requests detailed information for a specific title, rather than being done for every item shown on the home page. The reviews collected are processed using a pretrained RoBERTa-based sentiment model and categorized as positive, neutral, or negative. The application then calculates the percentage distribution of these categories. Redis is used to store sentiment results so that the same reviews do not need to be processed multiple times. The project does not try to access private Netflix information like individual user activity, viewing statistics, internal recommendation algorithms, or proprietary likes and dislikes.

1.3 Overview

CineSense follows a client-server structure where Next.js handles both the web interface and server-side application logic. The TMDB API serves as the main external data source for movie and television information, including titles, ratings, popularity, descriptions, search results, watch-provider details, and reviews. When users visit the home page, the application retrieves and presents relevant content without performing demanding sentiment analysis. When a user selects a specific title, the application fetches its detailed information and reviews. [3]

The reviews are then sent to a pretrained RoBERTa-based NLP sentiment model through the application's server-side processing layer. RoBERTa is a transformer-based language model designed as an improved method for BERT pretraining and has shown strong performance in various natural-language understanding tasks. The sentiment predictions for individual reviews are processed by the CineSense backend to calculate the overall percentage of positive, neutral, and negative reviews. The final sentiment result is stored in Redis, allowing later requests for the same title to access the existing result without going through the full analysis process again.

2. LITERATURE REVIEW

Digital movie and television information systems have changed how users find and assess entertainment content. Modern platforms provide details like title, genre, cast, release date, popularity, ratings, descriptions, and user reviews. However, traditional systems rely heavily on numerical ratings, which give a clear number but fail to explain why viewers liked or disliked a specific title. Audience reviews include important details, such as praising acting while criticizing storylines, which standard scores miss.

CineSense addresses this gap by combining movie metadata with natural language processing (NLP) to analyze audience reviews. Using a pretrained RoBERTa model for sentiment analysis, the platform automatically processes complex viewer feedback. Instead of making users read hundreds of reviews or rely solely on an isolated score, CineSense categorizes review text into positive, neutral, and negative opinions to create an overall sentiment breakdown.

[1] [4]

The system integrates this sentiment breakdown directly into a single web-based discovery interface. Powered by an external entertainment data API and server-side web development, along with Redis caching to prevent redundant processing, CineSense shows how NLP and efficient web infrastructure can work together to provide a deeper understanding of audience reception.

3. OVERALL DESCRIPTION

The proposed system is a public, web-based content-discovery and feedback platform focused on movies and web series. Unlike conventional review platforms, the system requires no account creation or login for browsing, rating, liking/disliking, or commenting — any visitor can interact with the catalog immediately. The title catalog itself is never entered manually; it is kept current through automatic synchronization with an external catalog API. User-generated signals — star ratings, likes/dislikes, and free-text comments — are continuously processed by a Machine Learning (ML) sentiment-analysis pipeline that produces a live sentiment score displayed on each title's dashboard. A single administrative role exists purely for moderation and system oversight (reviewing flagged comments, monitoring catalog-sync health, and monitoring model performance) and is the only part of the system gated behind a login.

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3.1 Product Perspective

The platform is a new, self-contained, dual-surface application comprising a Public surface (no account required) and a single Admin surface, backed by a shared database of titles, votes, comments, and computed sentiment scores. It integrates with an external catalog API for automatic title ingestion, and with an in-house ML/NLP pipeline for comment moderation, text cleaning, and sentiment classification. There is no payment gateway and no patient/doctor-style role separation — the defining design choice is that ordinary visitors never authenticate; only the operations team does, through the Admin module, purely to moderate content and monitor system health.

3.2 Product Function

- **F1 → Public Website Access and Browsing:** Loads the home page with trending titles and the public dashboard; no signup or login is required for any visitor.
- **F2 → Automated Catalog Sync via External API:** Keeps the title catalog current by automatically inserting, updating, and removing titles from a scheduled external API feed.
- **F3 → Search and Filter Titles:** Lets a visitor search by keyword/genre/language and sort results by popularity, rating, recency, or sentiment score.
- **F4 → Title Details and Sentiment Dashboard View:** Displays a title's metadata alongside its live rating, like/dislike, comment-count, and sentiment-score dashboard.
- **F5 → User Rating / Like / Dislike Submission:** Lets any visitor rate a title (1–5 stars) or like/dislike it, one submission per session.
- **F6 → User Comment Submission:** Lets any visitor post a free-text comment on a title, with an optional display name.
- **F7 → Comment Moderation and Spam Filtering:** Automatically screens every new comment for profanity, spam, or bot behaviour before it is shown publicly.
- **F8 → Comment Text Preprocessing and Cleaning:** Cleans, tokenizes, and normalizes approved comment text in preparation for the sentiment model.
- **F9 → Sentiment Analysis on Comments and Feedback:** Runs the cleaned comment text through the trained ML model to produce a Positive/Negative/Neutral label.
- **F10 → Aggregated Sentiment Score Computation:** Combines a title's star ratings, like/dislike ratio, and comment sentiment into one overall sentiment score.
- **F11 → Trending and Top-Rated Recommendations:** Ranks titles by sentiment score, popularity, and recency to populate the home page's highlight sections.
- **F12 → ML Model Monitoring and Retraining:** Periodically retrains the sentiment model on newly collected data and promotes it only if it outperforms the current one.
- **F13 → Admin Monitoring and Content Oversight:** The platform's only login-gated area, used to review flagged comments, sync logs, and model performance.

3.3 User Characteristics

- **Public Visitor:** Any internet user with basic digital literacy; no account needed to browse titles, view sentiment dashboards, search/filter, rate, like/dislike, or comment.
- **Admin (Moderator / Operator):** Technically proficient platform operator; logs in to review flagged comments, monitor catalog-sync logs, monitor ML model performance, and manage reported content.
- **Automated Actors (non-human):** The external catalog API (drives F2) and the scheduled ML retraining job (drives F12) act as system-level actors rather than end users.

3.4 Assumptions and Dependencies

- Visitors have reliable internet access on a computer, tablet, or smartphone; no installation is required.
- The external catalog API remains available and continues to provide a consistent schema for titles, metadata, posters, and cast.
- A meaningful sentiment score requires a minimum volume of ratings/votes/comments per title; titles with very little engagement may show 'Not Enough Data'.
- Comment moderation rules (profanity/spam/bot detection) are maintained and updated as new abuse patterns emerge.
- The ML sentiment model requires periodic retraining as new comments and ratings accumulate, and a human (Admin) reviews model-performance alerts.
- Because no login exists for the public, abuse-prevention (rate limiting, one-vote-per-session) relies on session/IP-based controls rather than account-based controls.

4. Specific Requirements

4.1 External Interface Requirements

4.1.1 User Interface

- **Public Web/Mobile Portal:** home page, search, title details & sentiment dashboard, rating/like/dislike, comment submission — all without login.
- **Admin Web Console:** login, flagged-comment review queue, catalog-sync log viewer, ML model-performance dashboard, reported-content management.

4.1.2 Hardware Interface

- **RAM:** 8 GB (min) — higher than a typical CRUD app to accommodate in-memory ML inference.
- **SSD/HDD:** 100 GB (min) — for the title catalog, comment history, and versioned model artifacts.
- **Network:** 5 Mbps (min).
- **GPU:** optional — recommended if a transformer-based (BERT) sentiment model is used in production.

4.1.3 Software Interface

- **Frontend:** HTML, CSS, JavaScript with a framework such as React or Angular.
- **Backend:** Python (FastAPI/Django/Flask) or Node.js, exposing REST endpoints for the public and admin surfaces.
- **NLP / ML:** NLTK/spaCy for text preprocessing; scikit-learn or a fine-tuned Transformer (Hugging Face) for sentiment classification; VADER optionally used for fast rating-assisted auto-labeling.
- **Database:** PostgreSQL/MySQL or MongoDB, storing titles, votes, comments, sentiment scores, and model metadata.
- **Task Queue:** Celery/Redis (or equivalent) for asynchronous comment moderation, sentiment scoring, and scheduled catalog-sync/retraining jobs.
- **External Catalog API:** a third-party movie/web-series metadata API (e.g., TMDB-style), integrated via REST and/or webhook.

4.1.4 Communication Interface

The application shall be developed as a responsive web platform compatible with major operating systems (Windows, macOS, Linux, Android, iOS) and modern browsers (Google Chrome, Firefox, Microsoft Edge). All client-server communication shall occur over HTTPS/TLS.

- **Development Software:** Integrated Development Environments (IDEs) such as Visual Studio Code, and API-testing tools such as Postman.

4.2 Functional Requirements

Each functional requirement below corresponds to a numbered flowchart (F1–F13) already produced for this project, expressed as Input / Output / Description / Next (the requirement(s) the flow leads into).

F1 → Public Website Access and Browsing:

- **Input:** An HTTP request to the landing page (no login required).
- **Output:** The home page rendered with trending titles and the public dashboard, or 'Service Temporarily Unavailable' if the server/database cannot be reached.
- **Description:** Confirms server and database connectivity, then loads the home page with trending titles across Movies, Web Series, Documentaries, and New Releases. No account creation or authentication is required at any point.
- **Next:** F3 (search); F4 (title details); F11 (trending section).

F2 → Automated Catalog Sync via External API:

- **Input:** A scheduled trigger, or a webhook call, from the external catalog API.
- **Output:** 'Catalog Sync Completed Successfully' with titles inserted, updated, or removed accordingly, or a logged 'API Sync Failed' with a retry scheduled.
- **Description:** On a recurring schedule, fetches the latest catalog (titles, metadata, posters, cast) from the external API, compares it against existing records, then inserts new titles, updates changed metadata, removes delisted titles, and syncs poster/trailer assets — entirely without manual admin data entry.
- **Next:** F1 (refreshed catalog visible on the home page); F4 (refreshed title details).

F3 → Search and Filter Titles:

- **Input:** A search keyword, or a genre/language filter.
- **Output:** A list of matching titles sortable by the chosen criterion, or 'No Titles Found'.
- **Description:** Queries the titles database, applies the requested filters, and lets the visitor sort results by popularity, rating, release date, or sentiment score.
- **Next:** F4 (title details for a selected result).

F4 → Title Details and Sentiment Dashboard View:

- **Input:** A selected Title ID.
- **Output:** The title's metadata, poster, cast, and synopsis together with its live dashboard (average rating, like/dislike count, comment count, sentiment score), or 'Title Not Available'.
- **Description:** Fetches the title record and renders it alongside the combined engagement-and-sentiment dashboard, computed in F10.
- **Next:** F5 (rate/like/dislike); F6 (comment); F10 (source of the displayed sentiment score).

F5 → User Rating / Like / Dislike Submission:

- **Input:** A star rating (1–5), a like, or a dislike on a specific title.
- **Output:** 'Feedback Recorded' with updated vote counts, or 'Already Rated / Invalid Request' if a duplicate submission is detected for the same session.
- **Description:** Stores the rating/like/dislike against the title, enforcing a single active submission per browser session to limit trivial abuse (no login exists to enforce this per-account).
- **Next:** F10 (aggregated sentiment score computation).

F6 → User Comment Submission:

- **Input:** Free-text comment on a movie/series, with an optional display name.
- **Output:** 'Comment Posted Successfully', or 'Comment Cannot Be Empty'.
- **Description:** Accepts a non-empty comment from any visitor and submits it to the server for moderation before it can appear publicly.
- **Next:** F7 (comment moderation and spam filtering).

F7 → Comment Moderation and Spam Filtering:

- **Input:** A newly submitted comment.
- **Output:** The comment published publicly; or rejected with 'Comment Violates Guidelines'; or flagged and routed to the admin review queue.
- **Description:** Automatically screens every comment against profanity, spam, and bot-behaviour rules (or a moderation model) before allowing it to appear publicly; borderline cases are queued for a human decision rather than auto-rejected or auto-published.
- **Next:** F8 (preprocessing of an approved comment); F13 (admin resolves a flagged comment).

F8 → Comment Text Preprocessing and Cleaning:

- **Input:** A batch of approved comment text.
- **Output:** Cleaned tokens ready for the sentiment model, or 'Empty Comment Skipped' if nothing meaningful remains after cleaning.
- **Description:** Strips HTML tags, emojis, and special characters, lowercases the text, tokenizes it, removes stopwords, applies lemmatization, and corrects obvious spelling errors.
- **Next:** F9 (sentiment analysis).

F9 → Sentiment Analysis on Comments and Feedback:

- **Input:** Cleaned comment tokens.
- **Output:** A sentiment label (Positive / Negative / Neutral) with a confidence score, or 'Feature Extraction Failed'.
- **Description:** Vectorizes the cleaned text (TF-IDF or embeddings) and runs it through the currently deployed ML sentiment model to classify the comment.
- **Next:** F10 (aggregated sentiment score computation).

F10 → Aggregated Sentiment Score Computation:

- **Input:** A title's accumulated ratings, likes/dislikes, and comment sentiment labels.
- **Output:** An updated overall sentiment score and trend for the title, or 'Not Enough Data for Sentiment Score' when engagement volume is too low.
- **Description:** Combines weighted signals — average star rating, like/dislike ratio, and the distribution of comment sentiment — into a single, continuously updated sentiment score per title.
- **Next:** F4 (dashboard update); F11 (trending/top-rated ranking).

F11 → Trending and Top-Rated Recommendations:

- **Input:** A home page refresh trigger (on load or on a schedule).
- **Output:** Updated 'Top Rated', 'Most Discussed', 'Trending Now', and 'Recommended' sections, or 'Insufficient Data – Showing Latest Titles' as a fallback.
- **Description:** Ranks titles by sentiment score, popularity, and recency to populate the home page's highlight sections.
- **Next:** F4 (visitor clicks through to a highlighted title).

F12 → ML Model Monitoring and Retraining:

- **Input:** A scheduled retraining trigger, or a model performance-drop alert.
- **Output:** 'Deploy Updated Model to Production' if the retrained model outperforms the current one; otherwise 'Retain Existing Production Model'; or 'Insufficient New Data – Skip Retraining'.
- **Description:** Retrains the sentiment model on newly accumulated comment and rating data, evaluates it on accuracy, F1-score, precision/recall, and data drift, and promotes it to production only if it beats the current model on these metrics.
- **Next:** F9 (subsequent sentiment scoring uses the promoted model).

F13 → Admin Monitoring and Content Oversight:

- **Input:** Admin ID and password.
- **Output:** An authenticated session opening the oversight dashboard, or 'Invalid Admin Credentials'.
- **Description:** The platform's only login-gated area. Lets the Admin review flagged comments, monitor catalog-sync logs, view ML model performance, and manage reported content.
- **Next:** F2 (inspect sync logs); F7 (resolve flagged comments); F12 (view/trigger model monitoring).

4.3 Performance and Non-Functional Requirements

4.3.1 Security

- Rate limiting shall be applied to rating/like/dislike/comment submissions per session/IP, since no account system exists to gate abuse.
- All comment input shall be sanitized to prevent XSS and injection attacks before storage or display.
- Admin credentials and any stored operational secrets shall be encrypted; the Admin console shall be the only authenticated surface.
- No comment shall be publicly visible until it has passed moderation (F7).
- All traffic between client and server shall be encrypted using TLS.

4.3.2 Maintainability

- The codebase shall be modular by concern — catalog-sync service, feedback (rating/comment) service, NLP/ML pipeline, and the admin console — to allow each to be updated independently.
- The ML pipeline (preprocessing, vectorization, model) shall be version-controlled so a retrained model (F12) can be rolled back if it underperforms after deployment.
- Regular updates should be conducted to improve moderation rules and model accuracy as new abuse or language patterns emerge.

4.3.3 Performance Requirements

- Browsing, search, and dashboard requests should be processed within 3 seconds on average.
- A newly submitted comment should complete moderation and preprocessing (F7–F8) within a few seconds under normal load.
- Sentiment scoring of an individual comment (F9) should complete near-real-time, with batch processing available as a fallback under high load, within a defined SLA (e.g., 5 minutes).
- The scheduled catalog sync (F2) should complete within its allotted maintenance window without blocking public read traffic.

4.3.4 Reliability

- The system should target 99.5% uptime for the public browsing and dashboard surface.
- Catalog sync (F2) failures shall be logged and automatically retried rather than silently dropped.
- Sentiment-pipeline job failures (F8–F9) shall be logged and retried; a title's dashboard should gracefully fall back to raw rating/like counts if the sentiment service is temporarily unavailable, rather than failing outright.

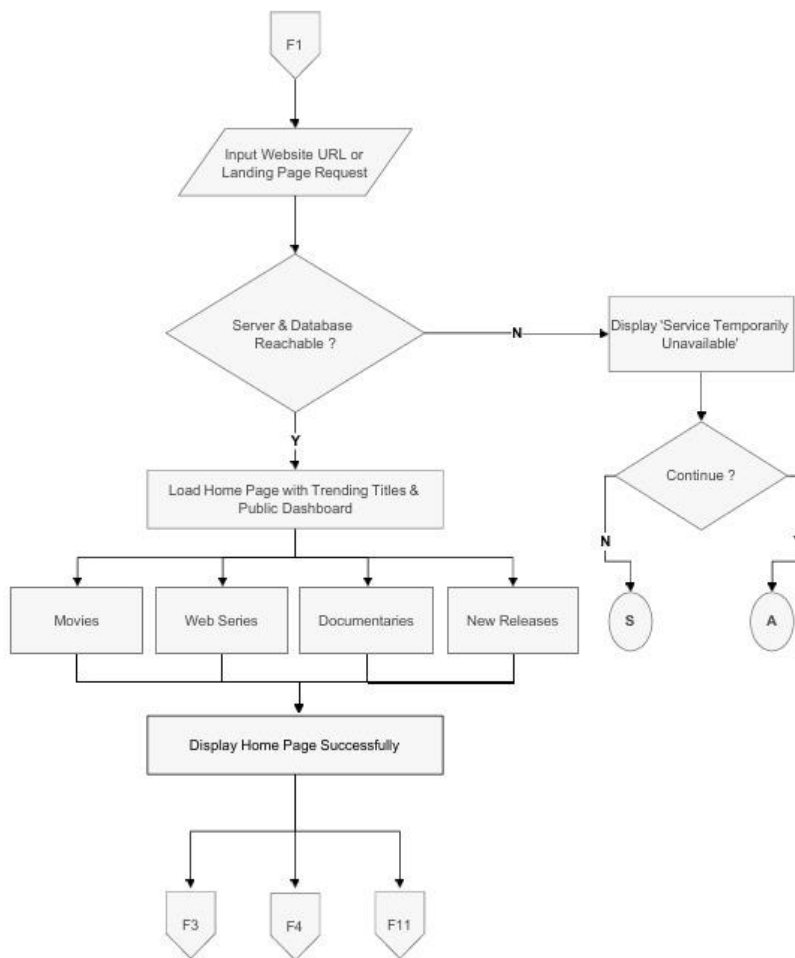
5. Conclusion

The Sentiment Analysis on Netflix Movies and Web Series platform is designed to give any visitor immediate, login-free access to an always-current catalog and a live, ML-driven read on how each title is actually being received — combining star ratings, likes/dislikes, and moderated comments into a single sentiment score. By automating catalog maintenance through an external API, screening user comments before publication, and continuously retraining the sentiment model on fresh engagement data, the system stays both current and trustworthy with minimal manual effort, while a single, narrowly scoped Admin role provides the oversight needed to keep content and model quality in check.

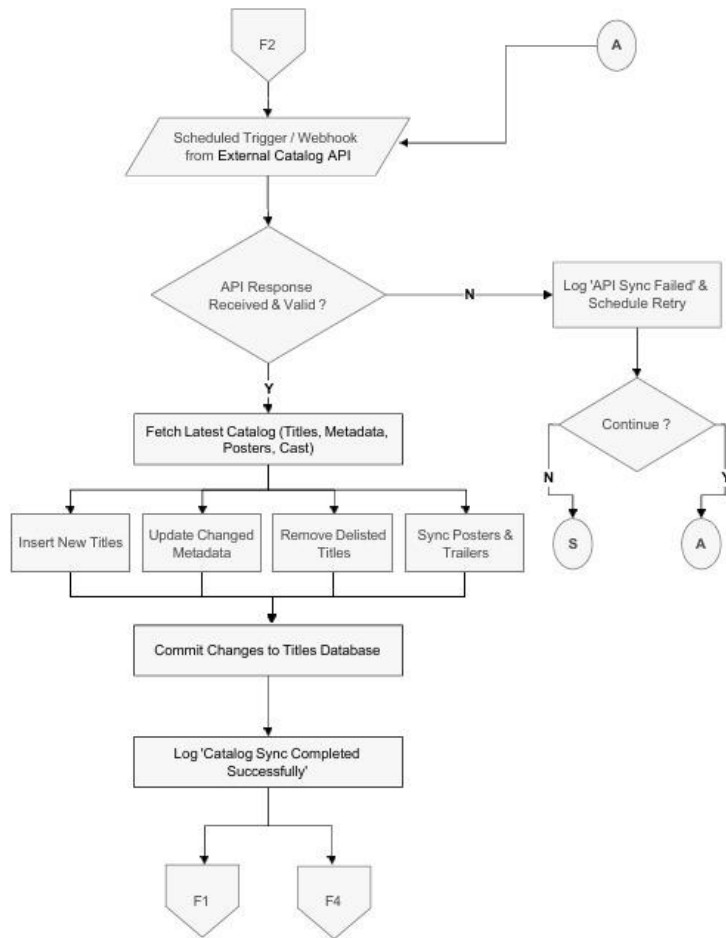
Open Items for Stakeholder Confirmation

- Whether a lightweight anti-abuse check (e.g., CAPTCHA, or stricter session/IP rate limiting) is required on comment and vote submission, given that no account system exists to gate abuse.
- Whether a visitor should be able to edit or delete their own comment afterward (e.g., via a short-lived session token), and for how long that window should remain open.
- The exact threshold/policy for automatically hiding a comment versus merely flagging it for admin review in F7.
- Data-retention period for anonymous engagement data (ratings, likes/dislikes, comments, and session/IP records used for rate-limiting), and the specific data-privacy regime applicable to the target market.
- Whether F5's 'per-session' rating limit is sufficient, or whether a more persistent (e.g., cookie- or device-based) limit is needed to reduce repeat voting after a session ends.

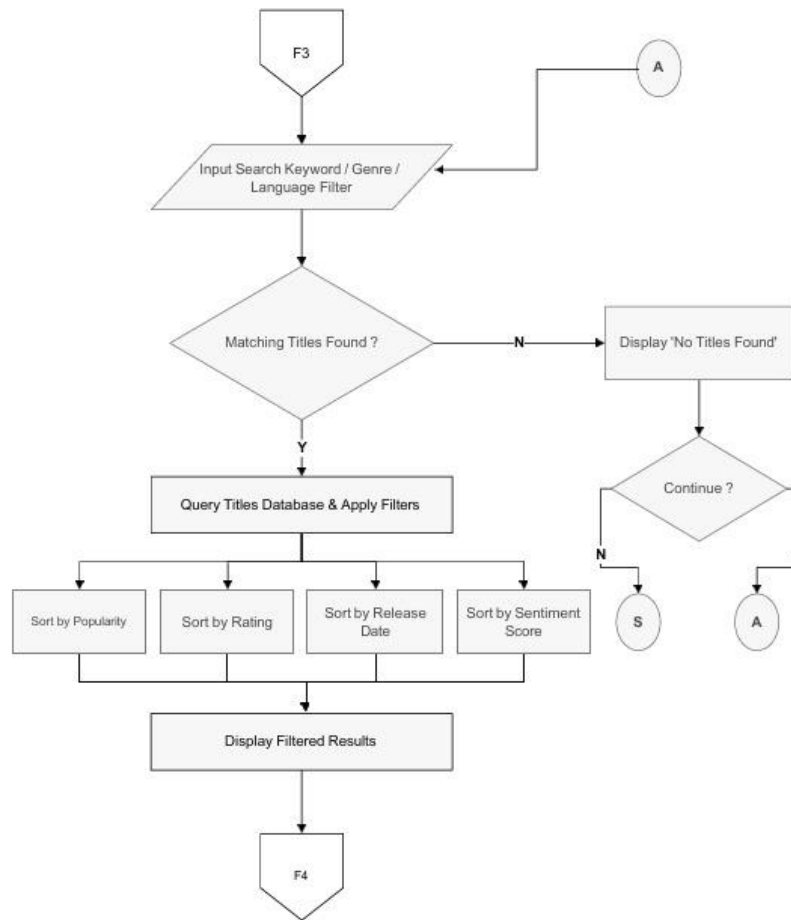
F1 : Public Website Access and Browsing



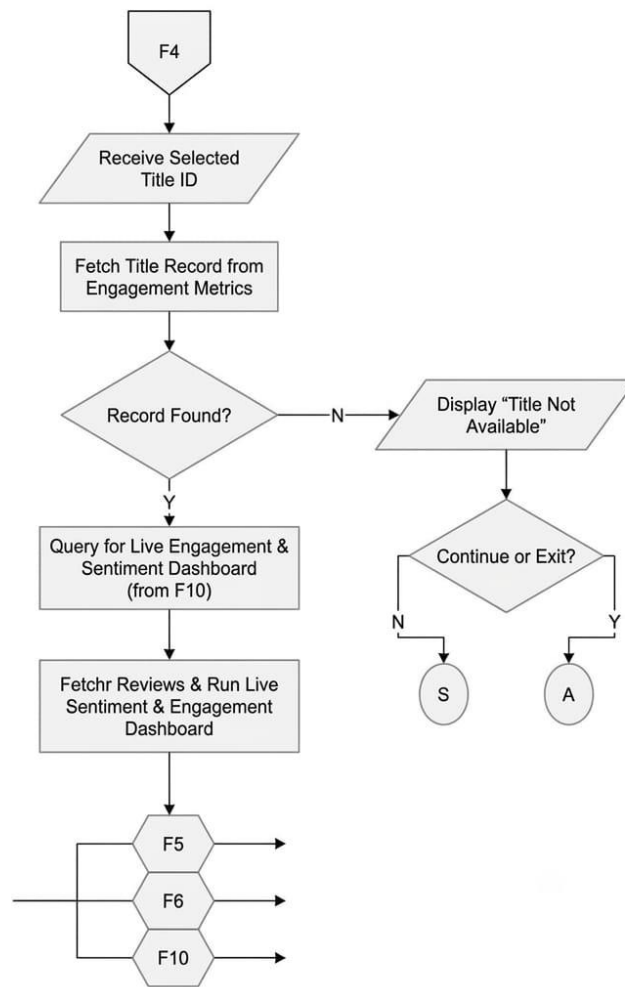
F2 : Automated Catalog Sync via External API



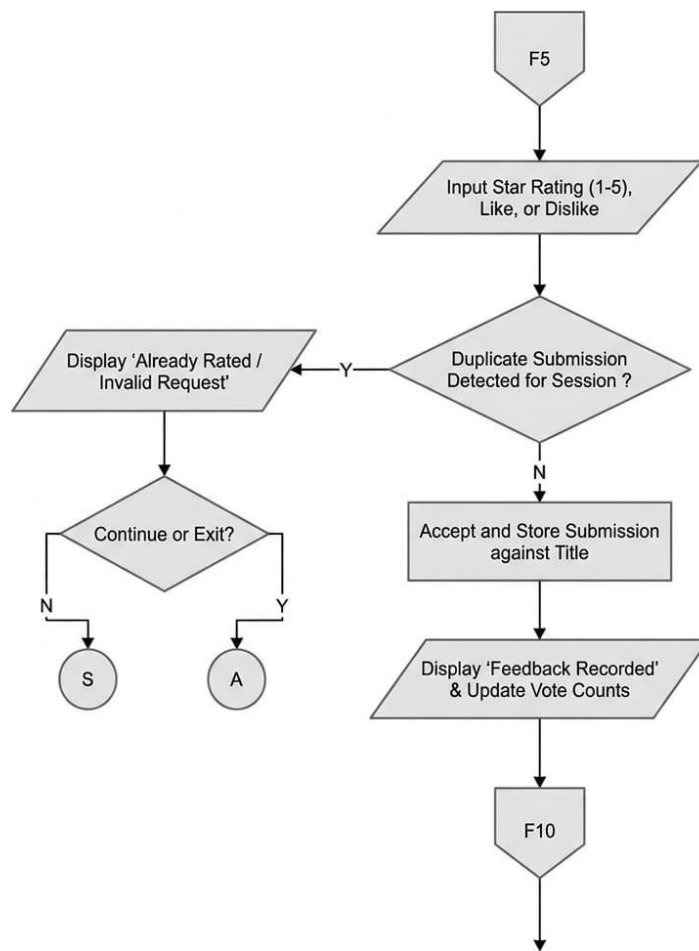
F3 : Search and Filter Titles



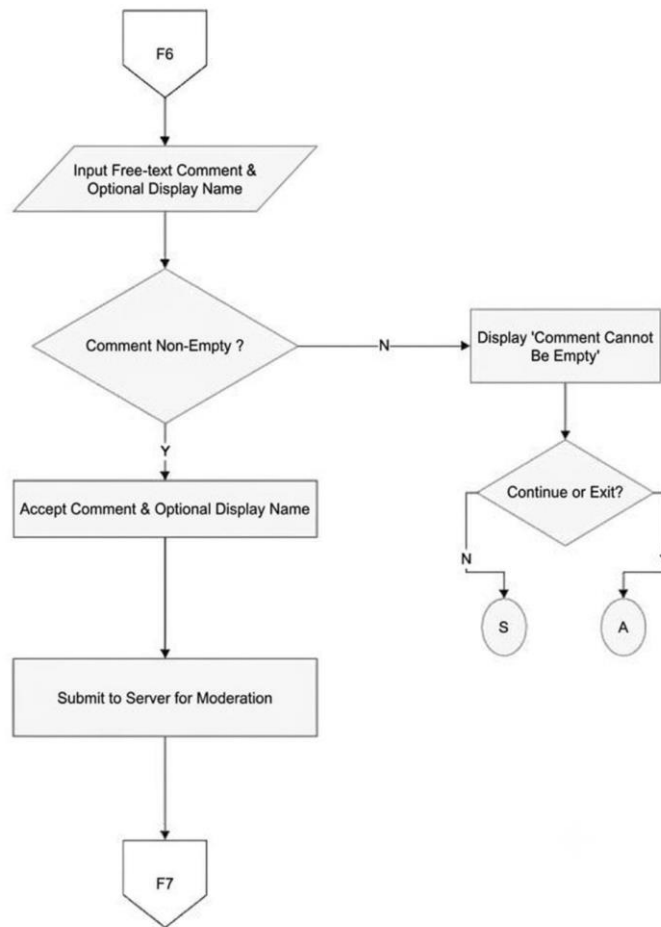
F4 : Title Details and Sentiment Dashboard View



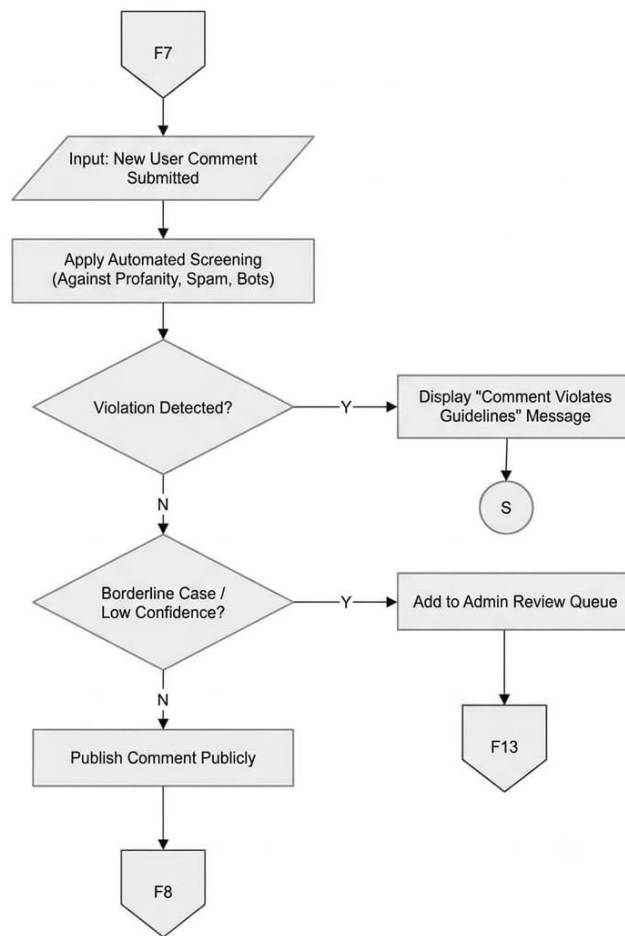
F5 : User Rating / Like / Dislike Submission

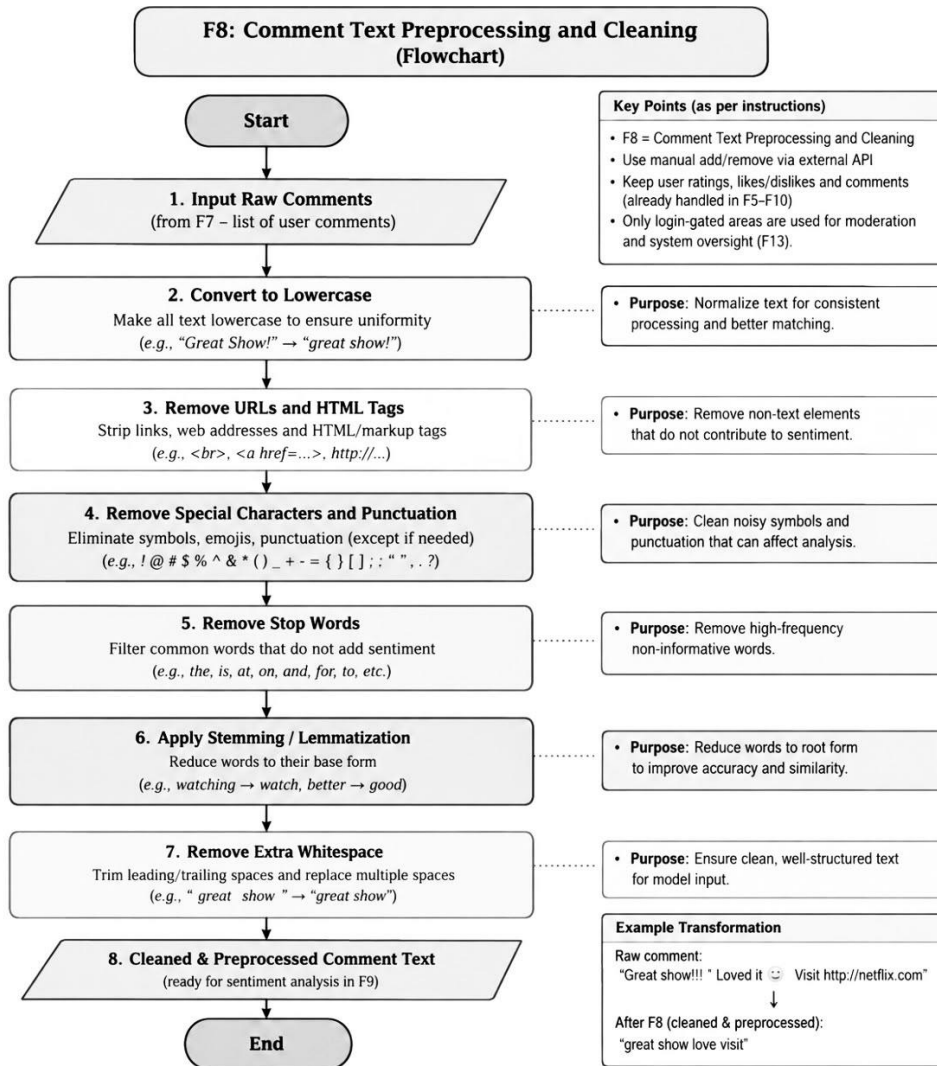


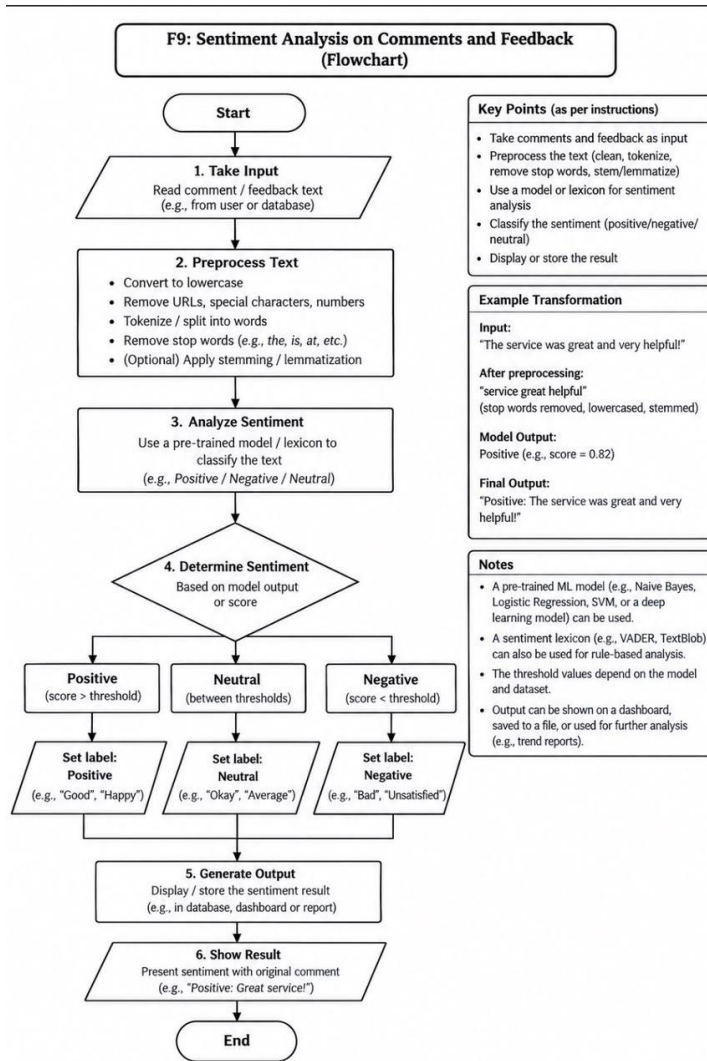
F6 : User Comment Submission



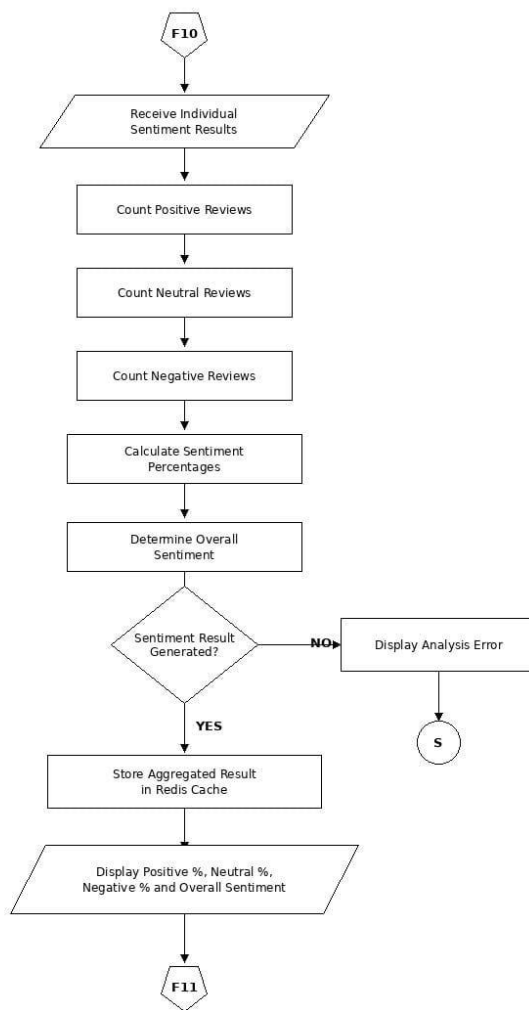
F7 : Comment Moderation and Spam Filtering



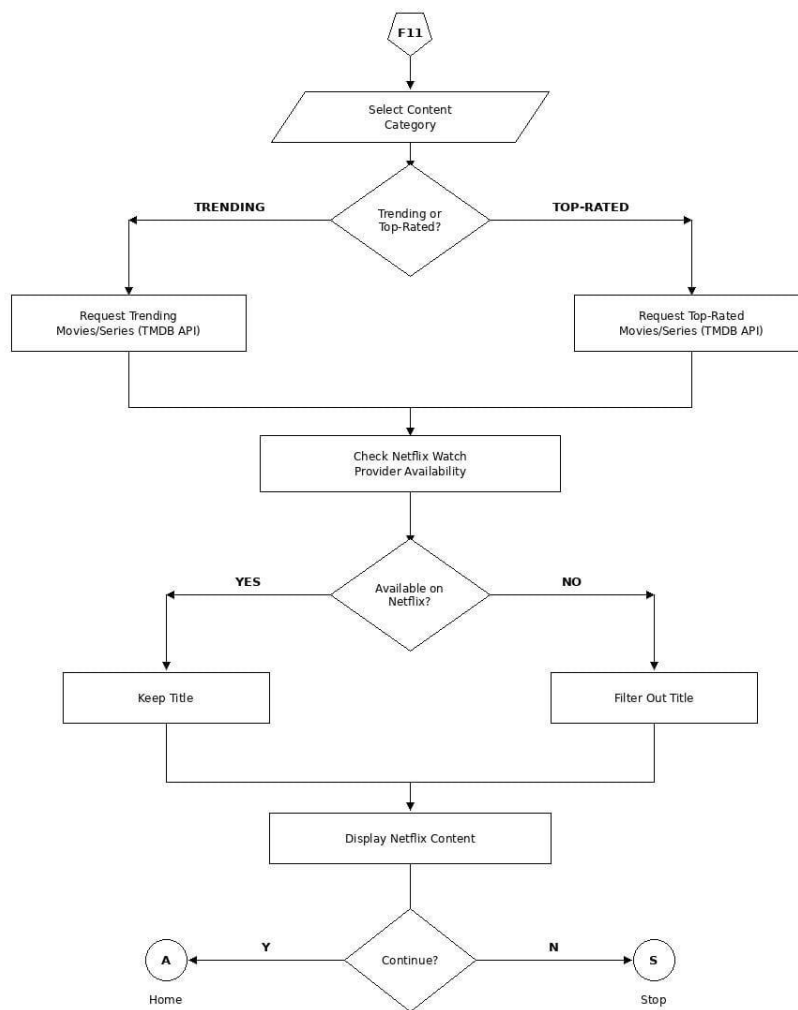




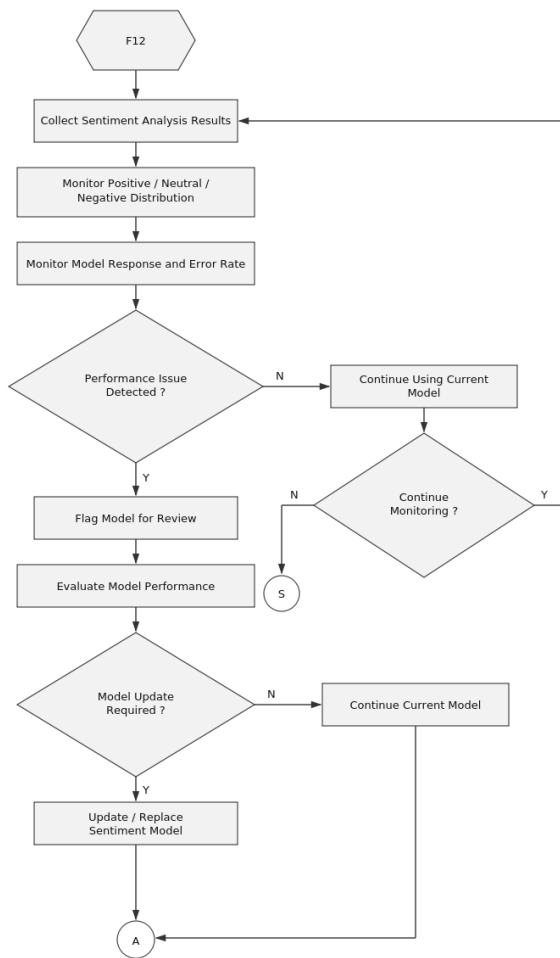
F10 : Aggregated Sentiment Score Computation



F11 : Trending and Top-Rated Content



F12 : CineSense - Sentiment Model Monitoring



F13 : CineSense - Admin Monitoring and Content Oversight

